MIL-I-25130B(AS)
7 October 1974
Superseding
MIL-I-25130A(WEP)
9 October 1961

MILITARY SPECIFICATION

INDICATOR, PRESSURE, OIL, 0-300PSI, TYPE MO-1

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of hermetically sealed, remote indicating, nonlighted, synchro operated oil pressure indicator. The type designation is MO-1.

2. APPLICABLE DOCUMENTS

*2.1 General. The following documents of the issue in effect on the date of the invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-7057	Indicators, Synchro, Aircraft, General Specification for
MIL-T-7748	Transmitters, Pressure, Synchro, Aircraft, 320-Degree Movement, High Temperature (177°C), General Purpose
MIL-N-18307	Nomenclature and Identification for Electronic Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment

STANDARDS

Federal

Fed. Std. No. 595	Colors

FSC 6620

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STANDARDS (Cont'd)

Military

MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS 28131	Transmitters, Pressure, Synchro, Aircraft, 320-Degree Movement, High Temperature (177°C), General Purpose
MS 33639	Case, Instrument, Clamp-mounted, Aircraft
MS 33678	Connector, Receptacle, Electric, Integral Mounting

2.2 Availability of documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

*3.1 Qualification. The indicator furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List (QPL) at the time set for opening of bids (see 4.3 and 6.4).

3.2 General requirements. The requirements of MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2.1 Requirement conflicts. In the event there are requirement conflicts between this specification and the general specification, this specification shall govern.

*3.3 Design and construction. The indicator shall be designed for use with a type MS 28131-12T or -27 transmitter conforming to MIL-T-7748.

*3.3.1 Case. The case shall be hermetically sealed. The case dimensions shall conform to Figure 1.

3.3.2 Pointer finish. The shaded portion of the pointer shall be finished with lusterless white conforming to Color No. 37875 of Fed. Std. No. 595.

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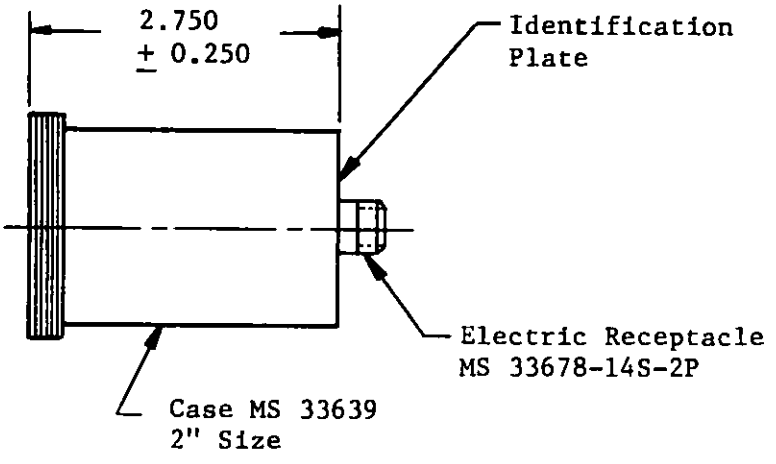


FIGURE 1. CASE

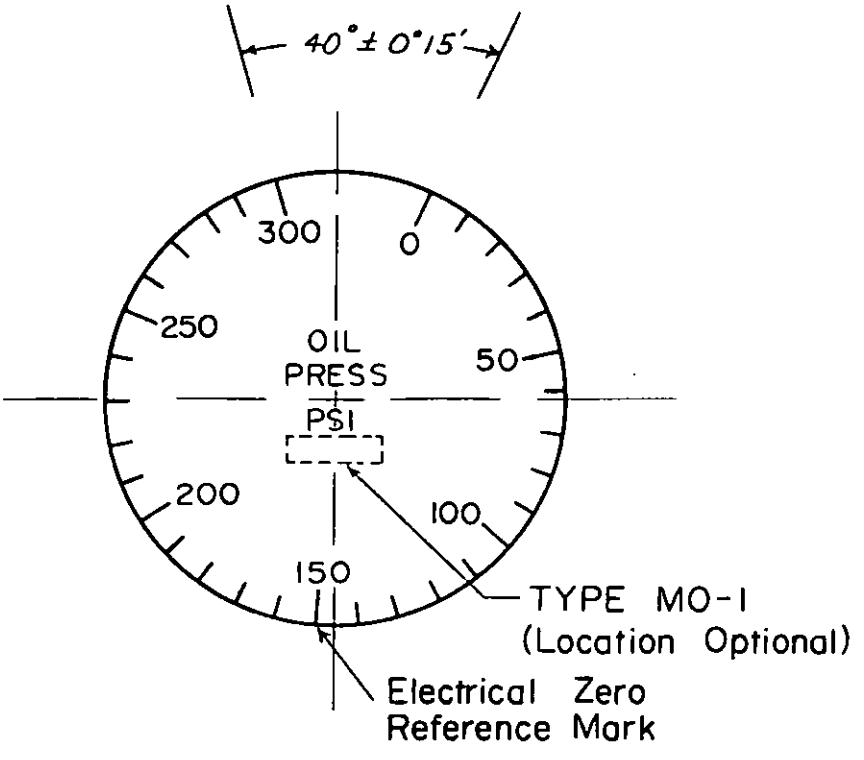


FIGURE 2. DIAL

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3.3.3 Dial. The dial shall conform to Figure 2. A minimum diameter of 1-3/4 inches shall be maintained across the outside ends of the graduations. The graduation for 230 pounds per square inch (psi) shall be located on the horizontal center line at the left side of the dial.

3.3.3.1 Dial markings. The dimensions of the markings shall be as follows:

Marking	Height or length (inch) ± 0.016	Width of line or graduation (inch) ± 0.005
Numerals 0, 50, 100, 150 200, 250, and 300	0.156	0.025
50-pound graduations	0.188	0.031
Graduation at 150 psi	0.140	0.031
10-pound graduations	0.094	0.020
Lettering "OIL PRESS"	0.125	0.025
Lettering "PSI"	0.125	0.025

The above markings shall be finished in lusterless white conforming to Color No. 37875 of Fed. Std. No. 595, and the dial background shall be finished in lusterless black conforming to Color No. 37038 of Fed. Std. No. 595.

3.3.3.1.1 "TYPE MO-1" marking. The "TYPE MO-1" marking shall be permanently and legibly marked on the dial in letters 0.063 inch high. These markings and all other markings not otherwise specified shall be finished in durable dull black.

3.3.4 Weight. The weight of the indicator shall not exceed 0.60 pound.

3.4 Identification of product. Units, assemblies, and parts shall be identified in accordance with MIL-N-18307 and MIL-I-7057.

*3.5 Reliability. The indicator shall have a specified mean time between failure (MTBF) of 1,500 hours when tested and accepted as specified in 4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance of the indicator shall be as specified in MIL-I-7057 and as specified herein. It shall include the qualification Inspection option of MIL-I-7057.

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*4.2 Test tolerances. The indicator shall perform within the tolerances specified in Tables I and II of this specification when tested in accordance with MIL-I-7057.

TABLE I

Scale Error Test Points and Tolerances

Test Points (psi)	Tolerances (± psi)		
	Room Temperature	Low Temperature	High Temperature
0	2.0	3.0	3.0
50	2.0	3.0	3.0
100	2.0	3.0	3.0
150	2.0	3.0	3.0
200	2.0	3.0	3.0
250	2.25	3.0	3.0
300	2.25	3.0	3.0

TABLE II

Tolerances of Other Tests

Tests	Tolerances (± psi)
Position error at 150 psi	2.0
Friction	
Room temperature	2.0
High temperature	2.0
Low temperature	3.0
Vibration	
Pointer oscillation	2.0
Pointer variation	2.0

*4.3 Qualification inspection. The qualification inspection shall consist of the reliability qualification phase test of 4.4.3 herein and all the tests specified in MIL-I-7057 in the order listed.

*4.3.1 Qualification inspection sample. The qualification inspection sample shall consist of a minimum of six indicators. These indicators shall have been previously subjected only to the individual tests. Three of the indicators shall be set aside. The remaining indicators shall

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be subjected to the reliability qualification phase test which will be conducted by the manufacturer at his plant. After the reliability qualification phase test has been successfully completed, the three set aside indicators shall be forwarded at the contractor's expense along with the individual test and reliability test reports to the laboratory designated by the qualifying activity for all the tests specified in MIL-I-7057.

*4.3.1.1 Qualification inspection sample identification. The qualification inspection samples shall be plainly identified by durable tags, securely attached and marked with the following information:

Sample for Qualification Inspection
Indicator, Pressure, Oil, 0-300PSI, Type MO-1
Submitted by (supplier's name, date, and part number) for Qualification Inspection in accordance with MIL-I-25130B(AS) under authorization (reference letter authorizing tests).

*4.3.2 Periodic qualification re-evaluation. In order to retain qualification of products approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification, on Form DD 1718, to the qualifying activity that his product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification shall be in two-year intervals from the date of original qualification.

*4.4 Quality conformance inspection. The supplier shall furnish all samples and shall be responsible for accomplishing all the inspections except the sampling plan B inspection, which will be conducted at a Government laboratory designated by the procuring activity. Quality conformance inspections, except for sampling plan B, shall be under the supervision of the Government quality control representative. The supplier shall furnish test reports showing detailed quantitative results for all tests required of the supplier, signed by an authorized representative of the supplier. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. The quality conformance inspection shall consist of the following tests:

(a) Individual
(b) Sampling plan A
(c) Sampling plan B
(d) Reliability production acceptance (sampling) phase

*4.4.1 Individual inspection. The individual inspection shall consist of the individual tests of MIL-I-7057 and the reliability production acceptance (sampling) phase test specified in 4.4.3.2. Each indicator accepted on the contract or order shall have conformed to all of the individual tests and shall be from a lot conforming to the reliability test.

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*4.4.2 Sampling plans. The sampling plans shall consist of sampling plans A and B. Indicators selected for the sampling plan tests shall first have passed the individual tests. Indicators which have been subjected to sampling plan A tests shall not be delivered on contract until they have been refurbished, if necessary, and resubmitted and passed all of the individual tests. Indicators which have been subjected to the sampling plan B tests shall not be delivered on contract.

*4.4.2.1 Sampling plan A sample selection. Indicators being submitted for sampling plan A testing shall be selected at random in accordance with the following schedule:

Quantity offered for acceptance	Quantity to be selected for inspection
First 15	(see note)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

NOTE: When sampling plan B is invoked, the quantity shall be zero. When sampling plan B is omitted, the quantity shall be one.

When an indicator failure is detected, no indicators from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected, and until the corrective action has been approved by the procuring activity. Also, when a failure occurs, the sampling procedure shall start a new cycle; that is, one indicator out of 15, then one indicator out of the next 50, etc.

*4.4.2.1.1 Sampling plan A inspection. Each indicator selected for sampling plan A shall be subjected to the sampling plan A tests specified in MIL-I-7057.

*4.4.2.2 Sampling plan B instructions. Three indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These indicators shall be forwarded at the supplier's expense to a government laboratory designated by the procuring activity. Each indicator shall be plainly identified by a durable tag, securely attached and marked with the following information:

Indicator, Pressure, Oil, 0-300PSI, Type MO-1
Submitted by (manufacturer's name and date)
for Sampling Plan B Inspection of
MIL-I-25130B(AS) in accordance with
Contract/Order No._____
Manufacturer's Part No._____

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*4.4.2.2.1 Sampling plan B approval. Approval of sampling plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material, or performance defect made evident during this test shall be corrected by the supplier to the satisfaction of the procuring activity. Failure of an indicator to pass any of the tests shall be cause for deliveries of indicators under the contract to cease until proper corrective action is approved and accomplished.

*4.4.2.2.2 Sampling plan B inspection. Each indicator selected for sampling plan B inspection shall be subjected to the sampling plan B tests specified in MIL-I-7057.

*4.4.3 Reliability assurance tests. The reliability assurance tests shall consist of the reliability qualification phase test and the reliability production acceptance (sampling) phase test in accordance with MIL-STD-781. Indicators selected for the reliability assurance tests shall first have passed the individual tests.

*4.4.3.1 Reliability qualification phase test. A minimum of three and a maximum of six indicators shall be tested as outlined in MIL-STD-781 under the section entitled "Qualification (Demonstration) Phase of Production Reliability Tests." Test Level F and Test Plan I shall be used.

*4.4.3.2 Reliability production acceptance (Sampling) phase test. The indicators shall be tested as outlined in MIL-STD-781 under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests." Test Level F and Test Plan II shall be used.

*4.4.3.2.1 Procedure for production acceptance (Sampling) phase. This test shall be conducted on each lot (see 6.3.2) during the life of the contract. It shall not start until the qualification phase test program has been completed. Testing on each lot shall begin no later than one week after the final indicator of the lot is produced. The number of indicators tested on each lot shall be not less than three nor more than seven. The indicators shall be tested until an accept or reject decision is reached. The test results of each lot shall be summarized for the procuring activity as soon as testing is completed on the lot. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the supplier's efforts to improve the equipment, the equipment quality control, etc.

*4.4.3.3 Test procedures and test details. Test procedures to be used for each phase of the reliability assurance tests shall be prepared by the supplier. Each procedure shall be submitted to and approved by the procuring activity before that phase of the reliability assurance testing contained in the procedure may begin. The test procedures shall contain all pertinent test details such as the length of the duty cycle, length of the heat portion of the cycle, performance characteristics to be measured, special failure criteria, etc. The following paragraphs shall be considered as minimum requirements and shall apply to both phases.

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*4.4.3.3.1 Duty cycle. The duty cycle shall be continuous. The duty cycle shall consist of cycling the indicator between 50 and 250 psi and return at a rate of 12 ± 1 cycles per hour. The power shall be "off" 3 to 5 minutes of each hour.

*4.4.3.3.2 Performance characteristics. The scale error and friction test shall be conducted at room temperature in accordance with MIL-I-7057 at least once each week. The scale error shall be within the tolerances of Table I. The friction error shall be within the room temperature tolerance of Table II.

*4.4.3.3.3 Failure criteria. Whenever performance characteristics fall below the requirement of 4.4.3.3.2 above, at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless the procuring activity and the supplier agree that one part caused the other parts to fail.

*4.4.3.3.4 Preventive maintenance. No preventive maintenance may be accomplished on the indicators while they are on test.

*4.4.3.3.5 Operational stability. No adjustment of any controls inaccessible to the operator shall be made during the reliability tests.

*4.4.3.4 Disposition of indicators upon completion of tests. Any indicator used for the reliability assurance tests may be delivered on contract provided it meets all of the following requirements:

(a) It is representative of production units currently being accepted.

(b) It is in "good as new" condition or has been refurbished (see 6.3.1, 6.3.3).

(c) It is otherwise satisfactory.

5. PREPARATION FOR DELIVERY

*5.1 The indicator shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-STD-794. The nomenclature shall be as follows: Indicator, Pressure, Oil, 0-300PSI, Type MO-1.

5.2 Reinspection date. The reinspection date marking shall be as specified by the procuring activity.

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6. NOTES

6.1 Intended use. The MO-1 indicator is intended for use in conjunction with the MS 28131-12T or -27 transmitter to provide a cockpit presentation of the oil pressure of the aircraft engine.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Color of dial markings, if other than as specified (see 3.3.3.1).

(c) Level of preservation, packaging, packing, and marking desired (see 5.1). Reinspection date (see 5.2).

(d) Laboratory that will conduct sampling plan B inspection (see 4.4).

(e) Lot size (see 6.3.2).

*6.2.1 Contract data requirements.

Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement (Short Title)	Applicable DID
4.3.1	Reliability test reports	DI-R-2118 (use para 10.5)
4.4	Quality Conformance test reports	DI-T-2072 (use para 10.1.b citing MIL-I-25130B(AS))
4.4.3.2.1	Reliability test results summary	DI-R-2118 (use para 10.5)
4.4.3.3	Reliability test procedure	DI-R-2124

Such data will be delivered as described on approved (numbered) DID's (Data Item Description/DD Form 1664) when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into the applicable contract.

*6.3 Definitions. The following definitions shall supplement those in MIL-I-7057.

*6.3.1 Good as new. "Good as new" shall mean instruments operated less than 10 percent of the specified MTBF operation.

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*6.3.2 Lot size. A lot shall be defined as 2 months production or as defined in the contract.

6.3.3 Refurbished. Refurbished shall mean that the instrument has been completely overhauled with all component parts meeting current parts standards, and the overhauled instrument shall have been subjected to and met all the requirements of a new instrument.

*6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they might be eligible to be awarded contracts or orders for the products covered by the specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20361, and information pertaining to qualification of products may be obtained from that activity.

*6.5 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence applies:

(a) Contract. The contract shall have precedence over any specification.

(b) This specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

*6.6 Marginal indications. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability

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whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-I-25130B(AS) INDICATOR, PRESSURE, OIL, 0-300PSI, TYPE MO-1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. *RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES*

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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